

Competitive Exam Practice 10: Virtual Memory and Secondary Storage

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. Provenance tags distinguish verified past-year questions from original practice material.

- Q1** [*GATE CS 2004 Q47, verified real PYQ — GATEOverflow V3 p.92*] [**NAT, 2 marks**] Consider a system with a two-level paging scheme in which a regular memory access takes 150 ns and servicing a page fault takes 8 ms. An average instruction takes 100 ns of CPU time and requires two memory accesses. The TLB hit ratio is 90%, and the page-fault rate is one in every 10,000 instructions. What is the effective average instruction execution time, in ns? Assume that a TLB miss requires two page-table accesses and the final data/instruction access.
- Q2** [*Original, GATE-pattern*] [**NAT, 2 marks**] A machine has a 20-bit virtual address, 1 KiB pages, and 16 physical frames. The virtual address is 0xABCDE. If its virtual page maps to physical frame 9, give the physical address in decimal.
- Q3** [*Original, GATE-pattern*] [**NAT, 1 mark**] A TLB lookup takes 10 ns and a main-memory access takes 50 ns. On a TLB hit, one memory access is needed after the lookup; on a TLB miss, two page-table accesses and one final data access are needed after the lookup. If the TLB hit ratio is 90% and there are no page faults, what is the effective access time in ns?
- Q4** [*Original, GATE-pattern*] [**MCQ, 1 mark**] A segment-table entry has base address 12,000 and limit 600. Which statement is correct for logical address (s, 550) when the entry is valid?
- (A) It is invalid because the physical address exceeds the limit.
 - (B) It is valid and translates to physical address 12,550.
 - (C) It is valid and translates to physical address 12,000.
 - (D) It is invalid because segmentation uses fixed-size pages.
- Q5** [*Original, GATE-pattern*] [**NAT, 2 marks**] Free holes have sizes 100, 500, 200, 300, and 600 KiB, in that order. A process requests 212 KiB. What is the remaining size, in KiB, of the selected hole under best-fit allocation?
- Q6** [*Original, GATE-pattern*] [**NAT, 2 marks**] In a RAID 5 stripe, three surviving data blocks are $D_0 = 1010$, $D_1 = 0110$, and $D_2 = 1100$. Compute the 4-bit parity block $P = D_0 \oplus D_1 \oplus D_2$ and report P as a binary number.

Q7 [*Original, GATE-pattern*] [**NAT, 1 mark**] An HDD has average seek time 4 ms, rotational speed 7,200 rpm, transfer time 0.2 ms, and controller overhead 0.1 ms. Using average rotational latency of half a rotation, what is the approximate access time in ms?

Q8 [*Original, GATE-pattern*] [**MCQ, 1 mark**] Which pairing is correct?

- (A) Paging – external fragmentation is the defining cost.
- (B) Segmentation – all allocation units have equal size.
- (C) RAID 0 – striping without redundancy.
- (D) RAID 1 – distributed dual parity.